# SECTION 3

# EMERGENCY AND MALFUNCTION PROCEDURES

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# SECTION 3

# EMERGENCY AND MALFUNCTION PROCEDURES

## 3.1 GENERAL

This section contains the recommended procedures for coping with various types of emergencies, malfunctions and critical situations.

**WARNING** AFTER AN ACTUAL EMERGENCY OR MALFUNCTION MAKE AN ENTRY IN THE HELICOPTER LOGBOOK AND, WHEN NECESSARY, THE EFFECTED SYSTEM LOGBOOK (E.G. ENGINE LOGBOOK). MAINTENANCE ACTION MAY BE REQUIRED AND NECESSARY BEFORE THE NEXT FLIGHT.

For definitions of terms, abbreviations and symbols used in this section, refer to section 1.

### 3.1.1 Basic Rules

These procedures deal with common emergencies. However, they do not prevent the pilot from taking additional action necessary to recover the emergency situation.

Although the procedures contained in this section are considered the best available, the pilot's sound judgement is of paramount importance when confronted with an emergency.

To assist the pilot during an inflight emergency, three basic rules have been established:

1. Maintain aircraft control
2. Analyse the situation
3. Take proper action.

**NOTE** It is impossible to establish a predetermined set of instructions that would provide a ready-made decision applicable to all situations.

### 3.1.2 Memory Items

**Emergency procedure steps which shall be performed immediately without reference to either this manual or the pilot's checklist are written in boldface letters on a gray background (as shown here) and shall be committed to memory.**

Therefore, those emergency procedure steps appearing without boldface letters may be accomplished referring to either this manual or the pilot's checklist, and when time and situation permit.

### 3.1.3 Operating Condition

The following terms are used in emergency procedures to describe the operating condition of a system, subsystem, assembly or component:

Affected — Fails to operate in the normal or usual manner.

Normal — Operates in the normal or usual manner.

### 3.1.4 Urgency of Landing

**NOTE** The type of emergency and the emergency conditions combined with the pilot's analysis of the condition of the helicopter and his proficiency are of prime importance in determining the urgency of a landing.

The following terms are used to reflect the degree of urgency of an emergency landing:

LAND IMMEDIATELY

The urgency of landing is paramount. Primary consideration is to assure survival of the occupants. Landing in water, trees or other unsafe areas should be considered only as a last resort.

LAND AS SOON AS POSSIBLE

Land without delay at the nearest adequate site (i.e. open field) at which a safe approach and landing can be made.

LAND AS SOON AS PRACTICABLE

The landing site and duration of flight are at the discretion of the pilot. Extended flight beyond the nearest approved landing area where appropriate assistance can be expected is not recommended.

## 3.2 WARNING AND CAUTION LIGHTS

Most emergency situations will be indicated by either a **red warning light** or a **yellow caution light** coming on.

A **red warning light** on the WARNING PANEL or in pilot's view indicates an emergency condition requiring immediate corrective action. Additional audio signals may be triggered.

A **yellow caution light** on the instrument panel indicates a malfunction or failure conditions which do not require immediate crew action but the possible need for future corrective action.

It is always possible that a warning/caution light will unnecessarily come on. Whenever possible, check the light against its associated instrument to verify that an emergency condition has actually occurred.

Following is an alphabetical listing of the warning and caution light indications with the respective conditions, any further indications, and the emergency procedure.

### 3.2.1 Warning Light Indications

## WARNING LIGHT INDICATIONS

**BAT 60**

**Conditions/Indications**

Battery temperature high (above 60°C)

**Procedure**

- **O N   G R O U N D**

| | | | |
|---|---|---|---|
| **1.** | **Battery sw** | **–** | **Off** |
| 2. | Engines | – | Leave running in idle |

If warning light remains on for more than 5 minutes:

| | | | |
|---|---|---|---|
| 3. | Engines | – | Shut down |

**CAUTION** BATTERY MUST BE INSPECTED OR REPLACED PRIOR TO NEXT FLIGHT.

- **I N   F L I G H T**

| | | | |
|---|---|---|---|
| **1.** | **Battery sw** | **–** | **Off** |
| 2. | LAND AS SOON AS PRACTICABLE | | |

**CAUTION** BATTERY MUST BE INSPECTED OR REPLACED AT DESTINATION PRIOR TO NEXT FLIGHT.

## WARNING LIGHT INDICATIONS

**BAT 70**

**Conditions/Indications**

Battery overtemperature (above 70°C)

**Procedure**

- **O N  G R O U N D**

| | | |
|---|---|---|
| **1. Battery sw** | **–** | **Off** |
| 2. Engines | – | Shut down |

**CAUTION** BATTERY MUST BE INSPECTED OR REPLACED PRIOR TO NEXT FLIGHT.

- **I N  F L I G H T**

| | | |
|---|---|---|
| **1. Battery sw** | **–** | **Off** |
| **2. LAND AS SOON AS POSSIBLE** | | |

After landing:

| | | |
|---|---|---|
| 3. Engines | – | Leave running in idle |
| 4. Collective lever | – | Lock |
| 5. Cyclic stick | – | Centered |

**NOTE** Continue flight only if visual inspection reveals no indication of battery overheating. Leave battery off or disconnect battery.

**CAUTION** BATTERY MUST BE INSPECTED OR REPLACED AT DESTINATION PRIOR TO NEXT FLIGHT.

## WARNING LIGHT INDICATIONS

F or F

(Engine 1) (Engine 2)

**Conditions/Indications**

Overtemperature in engine compartment

**Procedure**

- **ON GROUND**

| | | | |
|---|---|---|---|
| **1.** | **Both EMERG FUEL VALVE sw** | **–** | **CLOSE** |
| **2.** | **Both FUEL PUMPS/SUPPLY TANK sw** | **–** | **Off** |
| **3.** | **Both power levers** | **–** | **OFF** |
| **4.** | **Battery sw** | **–** | **Off** |
| 5. | Passengers | – | Alert/Evacuate |
| 6. | Fire | – | Extinguish if possible |

- **IN FLIGHT**

| | | | |
|---|---|---|---|
| **1.** | **EMERG FUEL VALVE sw (affected engine)** | **–** | **CLOSE** |
| **2.** | **FUEL PUMPS/SUPPLY TANK sw (affected engine)** | **–** | **Off** |
| **3.** | **Power lever (affected engine)** | **–** | **OFF** |
| 4. | Passengers | – | Alert |
| **5.** | **LAND AS SOON AS POSSIBLE** | | |

After landing:

| | | | |
|---|---|---|---|
| 6. | Fire | – | Extinguish if possible |

## WARNING LIGHT INDICATIONS

**FILT 1** and/or **FILT 2**

**Conditions/Indications**

Engine fuel filter(s) clogged

**Procedure**

- **ONE WARNING LIGHT ON**

LAND AS SOON AS PRACTICABLE

Be prepared for single engine failure

- **BOTH WARNING LIGHTS ON**

**LAND AS SOON AS POSSIBLE**

Be prepared for double engine failure

## WARNING LIGHT INDICATIONS

**GEN 1** or **GEN 2**

**Conditions/Indications**

Generator failure

– Voltmeter indication normal (27 - 29 V)
– **GEN** warning light (affected generator) on

Generator overvoltage

– Voltmeter indication above normal (30 V or more)
– **GEN** warning light (normal generator) on

**NOTE** Normal generator is disconnected by reverse current of affected generator.

**Procedure**

- **GENERATOR FAILURE**

1. STARTER/GENERATOR sw (affected generator) – Off, then GEN ON

If **GEN** warning light remains on:

2. STARTER/GENERATOR sw (affected generator) – Off
3. CURRENT IND sw – Select normal generator
4. Ammeter and voltmeter – Monitor

- **GENERATOR OVERVOLTAGE**

1. STARTER/GENERATOR sw (affected generator) – Off
2. **GEN** warning light (affected generator) – On
3. **GEN** warning light (normal generator) – Off
4. CURRENT IND sw – Select normal generator
5. Ammeter and voltmeter – Monitor

**NOTE** One generator alone will provide sufficient power for normal services. Non-essential DC-bus loads are shed automatically.

## WARNING LIGHT INDICATIONS

**GEN 1** and **GEN 2**

**Conditions/Indications**

Both generators have failed or are disconnected from the power distribution circuit.

**Procedure**

1. Each STARTER/GENERATOR sw in turn – Off, then GEN ON

If both **GEN** warning lights remain on:

2. Both STARTER/GENERATOR sw – Off
3. CURRENT IND sw – BUS BAR (BATT)
4. Electrical consumption – Reduce
5. Ammeter and voltmeter – Monitor
6. LAND AS SOON AS PRACTICABLE

| Residual Battery Endurance | | | | | |
|---|---|---|---|---|---|
| Continuous load [A] | 15 | 20 | 25 | 30 | 40 |
| Time [min] | 60 | 45 | 35 | 30 | 22 |

**NOTE** Calculations are based on an assumed minimum battery capacity of 15 Ah. Times include 10-min landing light operation and 10-min radio transmitting.

**WARNING** A TOTAL ELECTRICAL FAILURE (LOSS OF BOTH GENERATORS AND ONBOARD BATTERY POWER) WILL LIMIT THE FUEL AVAILABLE TO THAT QUANTITY CONTAINED IN THE SUPPLY TANK AT THE TIME OF FAILURE.

## WARNING LIGHT INDICATIONS

***EFFECTIVITY*** *Hydraulic Systems without Positive Cross-over Indication*

**HY. BLOCK**

***EFFECTIVITY*** *Hydraulic Systems with Positive Cross-over Indication*

**HY. BLOCK** and **HYD 1**

***EFFECTIVITY*** *ALL*

**Conditions/Indications**

Servo jam in system 1 with automatic cross-over to system 2

– Increased force required to move the affected control axis

**Procedure**

**CAUTION**
- DO NOT OPERATE HYD TEST SWITCH IN FLIGHT.
- DURING ENGINE SHUTDOWN, BATTERY SHALL REMAIN ON UNTIL ROTOR COMES TO A COMPLETE STOP.

LAND AS SOON AS PRACTICABLE.

## WARNING LIGHT INDICATIONS

### HYD 1

**Conditions/Indications**

System 1 pressure loss with automatic cross-over to system 2

**Procedure**

**CAUTION**
- DO NOT OPERATE HYD TEST SWITCH IN FLIGHT
- DURING ENGINE SHUTDOWN, BATTERY SHALL REMAIN ON UNTIL ROTOR COMES TO A COMPLETE STOP.

LAND AS SOON AS PRACTICABLE

## WARNING LIGHT INDICATIONS

**HYD 2**

**Conditions/Indications**

System 2 pressure loss, system 1 retains power

**Procedure**

**CAUTION**
- DO NOT OPERATE HYD TEST SWITCH IN FLIGHT.
- AVOID ABRUPT MANEUVERS AND BANK ANGLES GREATER THAN 30°.

LAND AS SOON AS PRACTICABLE.

## WARNING LIGHT INDICATIONS

LIMIT

MAST MOMENT INDICATOR

**Conditions/Indications**

MM indicator between end of **yellow arc** and 95% of full deflection

– **LIMIT** warning light on, however, goes off when decreasing mast moment

MM indicator **95% of full deflection** exceeded

– **LIMIT** warning light on and remains on even when decreasing mast moment

**Procedure**

- **BETWEEN END OF YELLOW ARC AND 95% OF FULL DEFLECTION**

| | | |
|---|---|---|
| 1. | Control input | – Reduce |
| 2. | **LIMIT** warning light | – Check goes off |

- **95% OF FULL DEFLECTION**

| | | |
|---|---|---|
| 1. | Control input | – Reduce |
| 2. | MM indicator | – Maintain within green arc |
| 3. | LAND AS SOON AS PRACTICABLE | |

**NOTE** Even once the MM indicator exceeds 95% of full deflection (close to white radial) and the LIMIT warning light remains on the warning light can be reset by pushing the TEST button.

## WARNING LIGHT INDICATIONS

**LOW FUEL**

### Conditions/Indications

Failure of both main tank pumps, or
Main tank empty

– Supply tank fuel quantity below 60 kg (75 l)

### Procedure

| | | | |
|---|---|---|---|
| 1. | Supply tank fuel level | – | Check |
| 2. | Main tank fuel level | – | Check |
| 3. | Both fuel main pump circuit breakers | – | Check in |
| 4. | Both fuel main pump switches | – | Check on |

If **LOW FUEL** warning light remains on:

5. LAND WITHIN 10 MINUTES

## WARNING LIGHT INDICATIONS

| MAG PLUG 1 | or | MAG PLUG 2 |
|---|---|---|

**Conditions/Indications**

Metal particles detected in engine oil

**Procedure**

**WARNING** A **MAG PLUG** WARNING LIGHT COMING ON TOGETHER WITH EITHER A TORQUE FLUCTUATION OR LOSS OF OIL PRESSURE SIGNIFIES AN IMPENDING ENGINE FAILURE. PERFORM SINGLE ENGINE EMERGENCY SHUTDOWN OF AFFECTED ENGINE IMMEDIATELY.

- **O N  G R O U N D**

1. **Single engine emergency shutdown – Perform**

- **I N  F L I G H T**

1. **Single engine emergency shutdown – Perform**
2. LAND AS SOON AS PRACTICABLE

If OEI operation is <u>not</u> possible:

3. **LAND AS SOON AS POSSIBLE**

*EFFECTIVITY 0451-9999, and 0001-0450 After SB 60-37*

**NOTE** For procedures applicable to helicopters 0001-0450 Before SB 60-37 see para 3.6.5, Oil Cooler Fan Failure.

**WARNING LIGHT INDICATIONS**

| OIL COOL |
|---|

**Conditions/Indications**

Failure of oil cooler fan

**Procedure**

- **A T  H O V E R  ( I G E / O G E )**

1. Engine and transmission oil temperatures – Monitor

**2. LAND AS SOON AS POSSIBLE**

- **I N  F L I G H T**

**1. Engine and transmission oil temperatures – Monitor**

**2. Airspeed – Attain as high as possible with minimum power required**

3. LAND AS SOON AS PRACTICABLE

If any oil temperature nears limit.

**4. LAND AS SOON AS POSSIBLE**

**NOTE** Transmission oil temperature transient range of 105-120 °C has a 10-min limit.

*EFFECTIVITY ALL*

## WARNING LIGHT INDICATIONS

| RPM | flashing |
|---|---|

### Conditions/Indications

$N_1$ Split

– $N_1$ Indicators show difference of 12% or more
– Audio signal - beeping tone

$N_{Ro}$ Low

– $N_{Ro}$ 95% or less
– Audio signal - beeping tone

$N_{Ro}$ High

– $N_{Ro}$ 103% or more
– Audio signal - none

**NOTE** If a two-stage warning unit is installed, a steady tone will come on with $N_{Ro}$ 108% or more.

### Procedure

● $N_1$ SPLIT

| | | |
|---|---|---|
| **1. Collective lever** | **–** | **Adjust to OEI-limits or below** |
| **2. Engine condition** | **–** | **Analyze** |

**NOTE** **RPM** warning light may come on during beep trim operation. Try to match $N_1$ until warning light goes off.

If residual torque is available:

3. Affected engine – Leave running

If residual torque is not available:

| | | |
|---|---|---|
| **4. Single engine emergency shut-down** | **–** | **Perform** |

● $N_{Ro}$ LOW/$N_{Ro}$ HIGH

| | | |
|---|---|---|
| **1. Collective lever** | **–** | **Adjust as necessary to maintain $N_{Ro}$ within normal operating range** |

## WARNING LIGHT INDICATIONS

| T OIL | or | XMSN OIL |
|---|---|---|

(whichever is installed)

**Conditions/Indications**

Transmission oil **pressure** below minimum

– Transmission oil pressure indication 0.5 kp/$cm^2$ or less

Transmission oil **temperature** above maximum

– Transmission oil temperature indication 105 °C or more

**Procedure**

• **OIL PRESSURE BELOW MINIMUM**

| | |
|---|---|
| **1. LAND IMMEDIATELY** | |

**NOTE** Descend with minimum power.

• **OIL TEMPERATURE ABOVE MAXIMUM**

**NOTE** Transmission oil temperature transient range of 105 - 120 °C has a 10-min limit.

| | | |
|---|---|---|
| **1. Power** | – | **Reduce** |
| **2. Airspeed** | – | **Attain as high as possible with minimum power required** |
| **3. LAND AS SOON AS POSSIBLE** | | |

3.2.2 **Caution Light Indications**

***EFFECTIVITY*** *0001-0160*

**CAUTION LIGHT INDICATIONS**

**EPU ON**

***EFFECTIVITY*** *0161-9999*

**CAUTION LIGHT INDICATIONS**

**EPU
ON**

***EFFECTIVITY*** *ALL*

**Conditions/Indications**

External power is applied to the electrical distribution system

**NOTE** **EPU ON** caution light going off <u>does not</u> indicate that the EPU cable is disconnected.

**Procedure**

After EPU starts:

1. EPU cable – Disconnected
2. EPU access door – Closed

*EFFECTIVITY 0001-0160*

## CAUTION LIGHT INDICATIONS

| L | R |
|---|---|

*EFFECTIVITY 0161-9999*

## CAUTION LIGHT INDICATIONS

| |
|---|
| ANTI ICING 1 |
| ANTI ICING 2 |

*EFFECTIVITY ALL*

**Conditions/Indications**

Engine anti-icing system in operation

– TOT increases by approximately 15 °C

**Procedure**

No corrective action necessary.

ENGINE EMERGENCY CONDITIONS

## 3.3 ENGINE EMERGENCY CONDITIONS

### 3.3.1 Single Engine Failure - Hover IGE

**Conditions/Indications**

- **RPM** warning ($N_1$ SPLIT) on

Affected engine:

- **GEN** warning light on
- Instruments indicate power loss

**Procedure**

| | | | |
|---|---|---|---|
| **1.** | **Collective lever** | – | **Adjust to OEI-limits or below** |
| **2.** | **Landing attitude** | – | **Establish** |
| **3.** | **Collective lever** | – | **Raise as necessary to stop descent and cushion landing** |

After landing:

| | | | |
|---|---|---|---|
| **4.** | **Single engine emergency shut-down** | – | **Perform** |

## ENGINE EMERGENCY CONDITIONS

### 3.3.2 Single Engine Failure - Hover OGE

**Conditions/Indications**

– **RPM** warning ($N_1$ SPLIT) on

Affected engine:

– **GEN** warning light on
– Instruments indicate power loss

**Procedure**

| | |
|---|---|
| **1. Collective lever** | **– Adjust to OEI-limits or below** |
| **2. Airspeed** | **– Increase if possible** |

● **F O R C E D  L A N D I N G**

| | |
|---|---|
| **3. Landing attitude** | **– Establish** |
| **4. Collective lever** | **– Raise as necessary to stop descent and cushion landing** |

After landing:

| | |
|---|---|
| **5. Single engine emergency shut-down** | **– Perform** |

● **T R A N S I T I O N  T O  O E I - F L I G H T**

| | |
|---|---|
| **3. Airspeed** | **– 60 KIAS ($V_Y$)** |
| **4. Single engine emergency shut-down** | **– Perform** |

5. LAND AS SOON AS PRACTICABLE

## ENGINE EMERGENCY CONDITIONS

### 3.3.3 Single Engine Failure - Takeoff

**Conditions/Indications**

- **RPM** warning ($N_1$ SPLIT) on

Affected engine:

- **GEN** warning light on
- Instruments indicate power loss

**Procedure**

| | | |
|---|---|---|
| **1. Collective lever** | – | **Adjust to OEI-limits or below** |

- **R E J E C T E D   T A K E O F F**

| | | |
|---|---|---|
| **2. Landing attitude** | – | **Establish** |
| **3. Collective lever** | – | **Raise as necessary to stop descent and cushion landing** |

After landing:

| | | |
|---|---|---|
| **4. Single engine emergency shut-down** | – | **Perform** |

- **T R A N S I T I O N   T O   O E I - F L I G H T**

| | | |
|---|---|---|
| **2. Airspeed** | – | **60 KIAS ($V_Y$)** |
| **3. Single engine emergency shut-down** | – | **Perform** |

4. LAND AS SOON AS PRACTICABLE

## ENGINE EMERGENCY CONDITIONS

### 3.3.4 Single Engine Failure - Flight

**Conditions/Indications**

– **RPM** warning ($N_1$ SPLIT) on

Affected engine:

– **GEN** warning light on
– Instruments indicate power loss

**Procedure**

| | | | |
|---|---|---|---|
| **1.** | **Collective lever** | **–** | **Adjust to OEI-limits or below** |
| **2.** | **Airspeed** | **–** | **60 KIAS ($V_Y$)** |
| **3.** | **Single engine emergency shutdown** | **–** | **Perform** |
| 4. | LAND AS SOON AS PRACTICABLE | | |

## ENGINE EMERGENCY CONDITIONS

### 3.3.5 Single Engine Landing

**Conditions/Indications**

One engine inoperative (OEI)

**Procedure**

LANDING APPROACH

| | |
|---|---|
| 1. Airspeed | – 60 KIAS ($V_Y$) |
| 2. Shallow approach | – Establish |

ON FINAL

| | |
|---|---|
| 3. Airspeed | – 40 KIAS |
| 4. Rate of descent | – 300 fpm |

TOUCHDOWN

| | |
|---|---|
| 5. Airspeed | – Reduce, depending on power available |
| 6. Landing attitude | – Establish |
| 7. Collective lever | – Raise as necessary to stop descent and cushion landing |

## ENGINE EMERGENCY CONDITIONS

### 3.3.6 Single Engine Emergency Shutdown

**NOTE** Before performing an inflight single engine emergency shutdown, determine if the situation will allow for OEI flight

Make certain that:

- the controls of the affected engine are selected, and
- the collective lever is adjusted to maintain the normal engine within OEI-limits.

**Procedure**

| | | | |
|---|---|---|---|
| **1.** | **Power lever (affected engine)** | **–** | **IDLE, then OFF** |
| **2.** | **EMER FUEL VALVE switch (affected engine)** | **–** | **CLOSE** |
| **3.** | **FUEL PUMPS/SUPPLY TANK switch (affected engine)** | **–** | **Off** |

## ENGINE EMERGENCY CONDITIONS

### 3.3.7 Inflight Restart

**NOTE** An inflight restart may be attempted after a flameout or shutdown subject to the pilot's evaluation of the cause of flameout.

**Procedure**

**CAUTION** DO NOT ATTEMPT INFLIGHT RESTART IF CAUSE OF ENGINE FAILURE IS OBVIOUSLY MECHANICAL.

1. Collective lever friction – Adjust to maintain position of lever when released
2. Electrical consumption – Reduce
3. Power lever – OFF
4. STARTER/GENERATOR sw – Off
5. EMERG FUEL VALVE sw – OPEN
6. FUEL PUMPS/SUPPLY TANK sw – On
7. IGNITION sw – On
8. STARTER/GENERATOR sw – START and hold

**CAUTION** ENGINE GAS PRODUCER SECTION MAY LOCKUP AFTER AN INFLIGHT FLAMEOUT OR SHUTDOWN. THEREFORE, EXCEPT DURING AN EMERGENCY, INFLIGHT RESTART SHOULD NOT BE ATTEMPTED DURING THE TIME PERIOD BETWEEN ONE MINUTE AND TEN MINUTES AFTER FLAMEOUT OR SHUTDOWN.

At 15% $N_1$:

9. Power lever – IDLE
10. TOT and $N_1$ – Monitor closely, observe TOT peak
11. $N_2$ increase – Monitor
12. Engine oil pressure – Positive indication

At 58% $N_1$:

13. STARTER/GENERATOR sw – Release
14. IGNITION sw – Off
15. STARTER/GENERATOR sw – GEN ON
16. Power lever – FLIGHT
17. Electrical consumers – As required
18. LAND AS SOON AS PRACTICABLE

If restart is not successful:

19. Single engine emergency shutdown – Perform

## ENGINE EMERGENCY CONDITIONS

### 3.3.8 Engine Overspeed - Driveshaft Failure

**Conditions/Indications**

– **RPM** warning ($N_1$ SPLIT) on

Affected engine:

– Torque decreases to zero
– $N_1$ decreases
– $N_2$ increases above $N_{Ro}$

Normal engine:

– Torque, $N_1$ and TOT increase
– $N_{Ro}$ and $N_2$ decrease

**Procedure**

| | | | |
|---|---|---|---|
| **1.** | **Collective lever** | **–** | **Adjust to OEI-limits or below** |
| **2.** | **Affected engine** | **–** | **Identify** |
| **3.** | **Single engine emergency shutdown** | **–** | **Perform** |

4. LAND AS SOON AS PRACTICABLE

## ENGINE EMERGENCY CONDITIONS

### 3.3.9 Engine Overspeed - Governor Failure

**Conditions/Indications**

- **RPM** warning may come on
- $N_{Ro}$ and both $N_2$ increase

Affected engine:

- Torque, $N_1$ and TOT increase

Normal engine:

- Torque, $N_1$ and TOT decrease

**Procedure**

| | | |
|---|---|---|
| **1. Collective lever** | **–** | **Adjust to OEI-limits or below** |
| **2. Affected engine** | **–** | **Identify** |
| **3. Power lever (affected engine)** | **–** | **Move towards IDLE until $N_2$ and $N_{Ro}$ stabilize in normal range** |

4. LAND AS SOON AS PRACTICABLE

## ENGINE EMERGENCY CONDITIONS

### 3.3.10 Engine Underspeed - Governor Failure

**Conditions/Indications**

– **RPM** warning may come on

Affected engine:

– Torque, $N_1$ and TOT decrease
– $N_2$ may decrease

Normal engine:

– Torque, $N_1$ and TOT increase
– $N_{Ro}$ and $N_2$ decrease

**Procedure**

| | | |
|---|---|---|
| **1.** | **Collective lever** | **– Adjust to OEI-limits or below** |
| **2.** | **Affected engine** | **– Identify** |
| **3.** | **ENGINE TRIM switch** | **– Compensate $N_2$/$N_{Ro}$ drop** |

4. LAND AS SOON AS PRACTICABLE

If no residual torque (affected engine) is available:

| | | |
|---|---|---|
| **5.** | **Single engine emergency shut-down** | **– Perform** |

## ENGINE EMERGENCY CONDITIONS

### 3.3.11 Compressor Stall

**Conditions/Indications**

- Dull humming or popping sounds
- Torque and $N_1$ indications erratic
- TOT fluctuates
- Possible light yaw jerk

**Procedure**

| | | |
|---|---|---|
| **1. Collective lever** | **–** | **Adjust to OEI-limits or below** |
| **2. Affected engine** | **–** | **Identify** |
| **3. Power lever** | **–** | **Move slowly to IDLE** |

If compressor stall is still evident:

| | | |
|---|---|---|
| **4. Single engine emergency shut-down** | **–** | **Perform** |
| 5. LAND AS SOON AS PRACTICABLE | | |

If compressor stall ends:

| | | |
|---|---|---|
| 4. Power lever | – | Move slowly to FLIGHT |

**NOTE** Severe stalls can cause damage to engine and drive system components and must be handled as an emergency condition.

Stalls of less severe nature (one or two low Intensity pops) may permit continued operation of the engine at a reduced power level thus avoiding the condition that caused the compressor stall.

## ENGINE EMERGENCY CONDITIONS

### 3.3.12 Droop Compensation Failure

**Conditions/Indications**

- $N_{Ro}$ and both $N_2$ decrease when collective lever is raised
- $N_{Ro}$ and both $N_2$ increase when collective lever is lowered

**Procedure**

**NOTE** Avoid large collective lever changes.

1. Collective lever – Maintain $N_{Ro}$ within limits
2. LAND AS SOON AS PRACTICABLE

## ENGINE EMERGENCY CONDITIONS

### 3.3.13 Engine Oil Pressure Low

**Conditions/Indications**

Affected engine:

– Oil pressure indication 3.5 kp/cm$^2$ or less

**Procedure**

1. **Single engine emergency shutdown – Perform**
2. LAND AS SOON AS PRACTICABLE

## ENGINE EMERGENCY CONDITIONS

### 3.3.14 Engine Oil Temperature High

**Conditions/Indications**

Affected engine:

– Oil temperature indication 107 °C or more

**Procedure**

| | | |
|---|---|---|
| **1. Power (affected engine)** | **–** | **Reduce** |
| **2. Oil temperature (affected engine)** | **–** | **Monitor** |

3. LAND AS SOON AS PRACTICABLE

If oil temperature remains above limit:

| | | |
|---|---|---|
| **4. Single engine emergency shut-down** | **–** | **Perform** |

## ENGINE EMERGENCY CONDITIONS

### 3.3.15 Double Engine Failure - Hover IGE

**Conditions/Indications**

- Left yawing motion
- $N_{Ro}$ and both $N_2$ decrease
- **RPM** warning ($N_{Ro}$ LOW) on
- Both **GEN** warning lights on
- Engine instruments (both engines) indicate power loss

**Procedure**

| | | |
|---|---|---|
| **1. Right pedal** | – | **Apply as necessary to stop yaw** |
| **2. Landing attitude** | – | **Establish** |
| **3. Collective lever** | – | **Raise as necessary to cushion landing** |

## ENGINE EMERGENCY CONDITIONS

### 3.3.16 Double Engine Failure - Flight

**Conditions/Indications**

- Left yawing motion
- $N_{Ro}$ and both $N_2$ decrease
- **RPM** warning ($N_{Ro}$ LOW) on
- Both **GEN** warning lights on
- Engine instruments (both engines) indicate power loss

**Procedure**

| | | |
|---|---|---|
| **1. Autorotation** | **–** | **Perform** |

## ENGINE EMERGENCY CONDITIONS

### 3.3.17 Double Engine Emergency Shutdown

**Procedure**

- **O N   G R O U N D**

| | | | |
|---|---|---|---|
| 1. | Both power levers | – | OFF |
| 2. | Both EMERG FUEL VALVE sw | – | CLOSE |
| 3. | Both FUEL PUMPS/SUPPLY TANK sw | – | Off |
| 4. | Battery sw | – | Off |

- **I N   F L I G H T**

| | | | |
|---|---|---|---|
| 1. | Both power levers | – | OFF |
| 2. | Both EMERG FUEL VALVE sw | – | CLOSE |
| 3. | Both FUEL PUMPS/SUPPLY TANK sw | – | Off |

## ENGINE EMERGENCY CONDITIONS

### 3.3.18 Autorotation

**Procedure**

| | | |
|---|---|---|
| **1. Collective lever** | **–** | **Full down, maintain $N_{Ro}$ within limits** |
| **2. Airspeed** | **–** | **75 KIAS recommended** |

**NOTE** Max range airspeed .................................... 100 KIAS at 85% $N_{Ro}$
Min rate-of-descent airspeed .......................... 60 KIAS at 100% $N_{Ro}$

| | | |
|---|---|---|
| **3. Double engine emergency shut-down** | **–** | **Perform** |

AT APPROX 100-FT AGL:

| | | |
|---|---|---|
| 4. Flare attitude | – | Establish (approx 15° to 25°) to reduce forward speed and rate of descent; control $N_{Ro}$ |

AT APPROX 6 TO 10 FT AGL:

| | | |
|---|---|---|
| 5. Flare attitude | – | Reduce to approx 5° |
| 6. Heading | – | Maintain |
| 7. Collective lever | – | Raise to stop descent and cushion landing |
| **8. Battery switch** | **–** | **Off** |

**NOTE** The appropriate values must be adjusted according to prevailing conditions of gross mass, wind and terrain.

# FIRE EMERGENCY CONDITIONS

## 3.4 FIRE EMERGENCY CONDITIONS

### 3.4.1 Cabin/Cargo Compartment Fire

**Conditions/Indications**

– Smoke, burning odor, flames

**Procedure**

- **O N   G R O U N D**

| | | |
|---|---|---|
| **1. Passengers** | – | **Alert/Evacuate** |
| **2. Double engine emergency shut-down** | – | **Perform** |
| **3. Fire** | – | **Extinguish if possible** |

- **I N   F L I G H T**

| | | |
|---|---|---|
| **1. Passengers** | – | **Alert** |
| **2. Fire** | – | **Extinguish if possible** |
| **3. Toxic fumes, smoke** | – | **Eliminate: Open sliding doors, windows and vents** |
| **4. LAND AS SOON AS POSSIBLE** | | |

After landing:

| | | |
|---|---|---|
| **5. Double engine emergency shut-down** | – | **Perform** |

## FIRE EMERGENCY CONDITIONS

### 3.4.2 Electrical Fire

**Conditions/Indications**

– Odor of burning insulation and/or acrid smoke

**Procedure**

- **ON GROUND**

| | | | |
|---|---|---|---|
| **1.** | **Passengers** | **–** | **Alert/Evacuate** |
| **2.** | **Double engine emergency shut-down** | **–** | **Perform** |
| **3.** | **EPU, if connected** | **–** | **Disconnect** |
| **4.** | **Fire** | **–** | **Extinguish if possible** |

- **IN FLIGHT**

| | | | |
|---|---|---|---|
| **1.** | **Passengers** | **–** | **Alert** |
| **2.** | **Both STARTER/GENERATOR sw** | **–** | **Off** |
| **3.** | **Battery sw** | **–** | **Off** |
| **4.** | **Fire** | **–** | **Extinguish if possible** |
| 5. | Electrical consumption | – | Reduce |
| 6. | Battery sw | – | On |
| 7. | CURRENT IND sw | – | BUS BAR (BATT) |
| 8. | Ammeter and voltmeter | – | Monitor |
| 9. | LAND AS SOON AS PRACTICABLE | | |

| Residual Battery Endurance | | | | | |
|---|---|---|---|---|---|
| Continuous load [A] | 15 | 20 | 25 | 30 | 40 |
| Time [min] | 60 | 45 | 35 | 30 | 22 |

**NOTE** Calculations are based on an assumed minimum battery capacity of 15 Ah. Times include 10-min landing light operation and 10-min radio transmitting.

If indications of electrical fire continue:

| | | | |
|---|---|---|---|
| **10.** | **Battery sw** | **–** | **Off if flight conditions permit** |
| **11.** | **LAND AS SOON AS POSSIBLE** | | |

## 3.5 TAIL ROTOR FAILURE CONDITIONS

### 3.5.1 Tail Rotor Drive Failure - Hover

**Conditions/Indications**

Complete loss of tail rotor thrust

– Rapid uncontrolled right yawing motion

**Procedure**

● **H O V E R  I G E**

| | |
|---|---|
| **1. Collective lever** | **– Full down** |

and simultaneously

| | |
|---|---|
| **2. Landing attitude** | **– Establish** |

After landing:

| | |
|---|---|
| **3. Double engine emergency shut-down** | **– Perform** |

● **H O V E R  O G E**

| | |
|---|---|
| **1. Collective lever** | **– Full down** |

If height permits:

| | |
|---|---|
| 2. Airspeed | – Gain |
| **3. Double engine emergency shut-down** | **– Perform** |
| **4. Collective lever** | **– Raise to stop descent and cushion landing** |

## TAIL ROTOR FAILURE CONDITIONS

### 3.5.2 Tail Rotor Drive Failure - Flight

**Conditions/Indications**

Complete loss of tail rotor thrust which may be caused by a severed drive shaft and/or seperation of the tail rotor assembly from the helicopter.

– Rapid right yawing motion eventually combined with a right roll and downward pitch.

**Procedure**

**WARNING** EXTENDED FLIGHT IS NOT POSSIBLE AFTER TAIL ROTOR DRIVE SYSTEM FAILURE. AUTOROTATION MUST BE ENTERED IMMEDIATLY.

| | |
|---|---|
| **1. Collective lever** | **– Full down immediately, maintain $N_{RO}$ within limits** |
| **2. Airspeed** | **– 75 KIAS recommended** |

**NOTE** Airspeed indication may be unreliable due to right yaw flight condition.

| | |
|---|---|
| **3. Double engine emergency shut-down** | **– Perform** |

**WARNING** LANDING WITHOUT THE TAIL ROTOR OPERATING IS HAZARDOUS AS THE HELICOPTER WILL TOUCH DOWN IN AN EXTREME YAW ATTITUDE AND PROBABLY ROLL OVER.

| | |
|---|---|
| **4. Modified side flare** | **– As necessary to obtain zero groundspeed upon touchdown** |

**NOTE** The rate of descent will be higher than normal due to the extreme yaw attitude and will require a modified side flare to reduce the groundspeed to as near zero as possible to prevent the helicopter from rolling over upon touchdown.

| | |
|---|---|
| **5. Collective lever** | **– Raise to stop descent and cushion landing** |

## TAIL ROTOR FAILURE CONDITIONS

### 3.5.3 Fixed-pitch Tail Rotor Control Failures

**Conditions/Indications**

Fixed-pitch tail rotor failure

- No directional response after pedal movement
- Locked pedals

**Procedure**

The emergency procedures for a tail rotor control failure will vary depending on flight conditions, power setting and mass of the helicopter.

Lowering or raising the collective lever will result in a left or right yawing motion to a degree dependent upon the tail rotor thrust remaining. Additionally, at high airspeeds, an ever increasing yaw oscillation may occur, however, is dampened by reducing airspeed to approx 60 KIAS and, if possible, increasing the power setting.

Approach and landing should be made with a suitable rate of descent, controlling sideslip angle with the collective lever. If the situation permits, a crosswind landing should be performed to minimize the amount of sideslip. Before touchdown, the ground speed should be reduced to a minimum. Touchdown so that the skids are aligned with the direction of landing.

**Right Pedal Fixed-pitch Settings**

Raising collective lever will result in a right yawing motion. Lower collective lever to control sideslip and touch down with forward speed.

**Left Pedal Fixed-pitch Settings**

Lowering collective lever will result in a left yawing motion. Approach with a right sideslip and reduce ground speed to allow for raising the collective lever as necessary to touch-down.

## TAIL ROTOR FAILURE CONDITIONS

### 3.5.4 Pedal Vibrations

**Conditions/Indications**

Impending tail rotor system failure

– Unusual pedal vibrations

**Procedure**

**LAND AS SOON AS POSSIBLE**

## SYSTEM EMERGENCY/MALFUNCTION CONDITIONS

## 3.6 SYSTEM EMERGENCY/MALFUNCTION CONDITIONS

### 3.6.1 Electrical Short Circuit

**Conditions/Indications**

- Voltmeter indicates low voltage
- Ammeter may indicate excessive current
- Failure of equipment powered by affected bus
- **GEN 1** and/or **GEN 2** warning light may come on

**Procedure**

| | | |
|---|---|---|
| **1. Both STARTER/GENERATOR sw** | **–** | **Off** |
| 2. CURRENT IND sw | – | BUS BAR (BATT) |
| 3. Electrical consumption | – | Reduce |
| 4. Ammeter and voltmeter | – | Monitor |
| 5. LAND AS SOON AS PRACTICABLE | | |

| **Residual Battery Endurance** | | | | | |
|---|---|---|---|---|---|
| Continuous load [A] | 15 | 20 | 25 | 30 | 40 |
| Time [min] | 60 | 45 | 35 | 30 | 22 |

**NOTE** Calculations are based on an assumed minimum battery capacity of 15 Ah. Times include 10-min landing light operation and 10-min radio transmitting.

If electrical short circuit indications continue:

| | | |
|---|---|---|
| **6. Battery sw** | **–** | **Off if flight conditions permit** |
| **7. LAND AS SOON AS POSSIBLE** | | |

## SYSTEM EMERGENCY/MALFUNCTION CONDITIONS

### 3.6.2 Cyclic Trim Actuator Failure/Runaway

**Conditions/Indications**

– Unsymmetrical cyclic stick forces

**NOTE** Cyclic stick full travel remains available.

**Procedure**

**CAUTION** DURING ENGINE SHUTDOWN, CYCLIC STICK SHALL BE IN CENTERED POSITION UNTIL ROTOR COMES TO A COMPLETE STOP.

LAND AS SOON AS PRACTICABLE.

## SYSTEM EMERGENCY/MALFUNCTION CONDITIONS

### 3.6.3 Supply Tank Fuel Pump Failure

**Conditions/Indications**

Affected engine:

– Fuel pressure indication below 0.6 kp/cm$^2$

**Procedure**

| | | |
|---|---|---|
| 1. | FUEL PUMPS/SUPPLY TANK sw (affected engine) | – Off |
| 2. | Altitude | – 13.700 ft or less |
| 3. | Bank angles above 45° and abrupt maneuvers | – Avoid |
| 4. | LAND AS SOON AS PRACTICABLE | |

## SYSTEM EMERGENCY/MALFUNCTION CONDITIONS

### 3.6.4 Servo Jam - Hydraulic System 2

**Conditions/Indications**

– Increased force required to move the affected control axis

**Procedure**

**CAUTION**
- DO NOT OPERATE HYD TEST SWITCH IN FLIGHT.
- AVOID ABRUPT MANEUVERS AND BANK ANGLES GREATER THAN 30°.

LAND AS SOON AS PRACTICABLE

*EFFECTIVITY 0001-0450 Before SB 60-37*

## SYSTEM EMERGENCY/MALFUNCTION CONDITIONS

### 3.6.5 Oil Cooler Fan Failure

**Conditions/Indications**

Failure of oil cooler fan

– Rapid oil temperature increase of both engines followed by transmission oil temperature increase.

**Procedure**

- **AT HOVER (IGE/OGE)**

**1. LAND AS SOON AS POSSIBLE**

- **IN FLIGHT**

| | | |
|---|---|---|
| **1. Airspeed** | **–** | **Attain as high as possible with minimum power required** |

2. LAND AS SOON AS PRACTICABLE

If any oil temperature nears limit:

**3. LAND AS SOON AS POSSIBLE**

**NOTE** Transmission oil temperature transient range of 105-120 °C has a 10-min limit.

*EFFECTIVITY ALL*